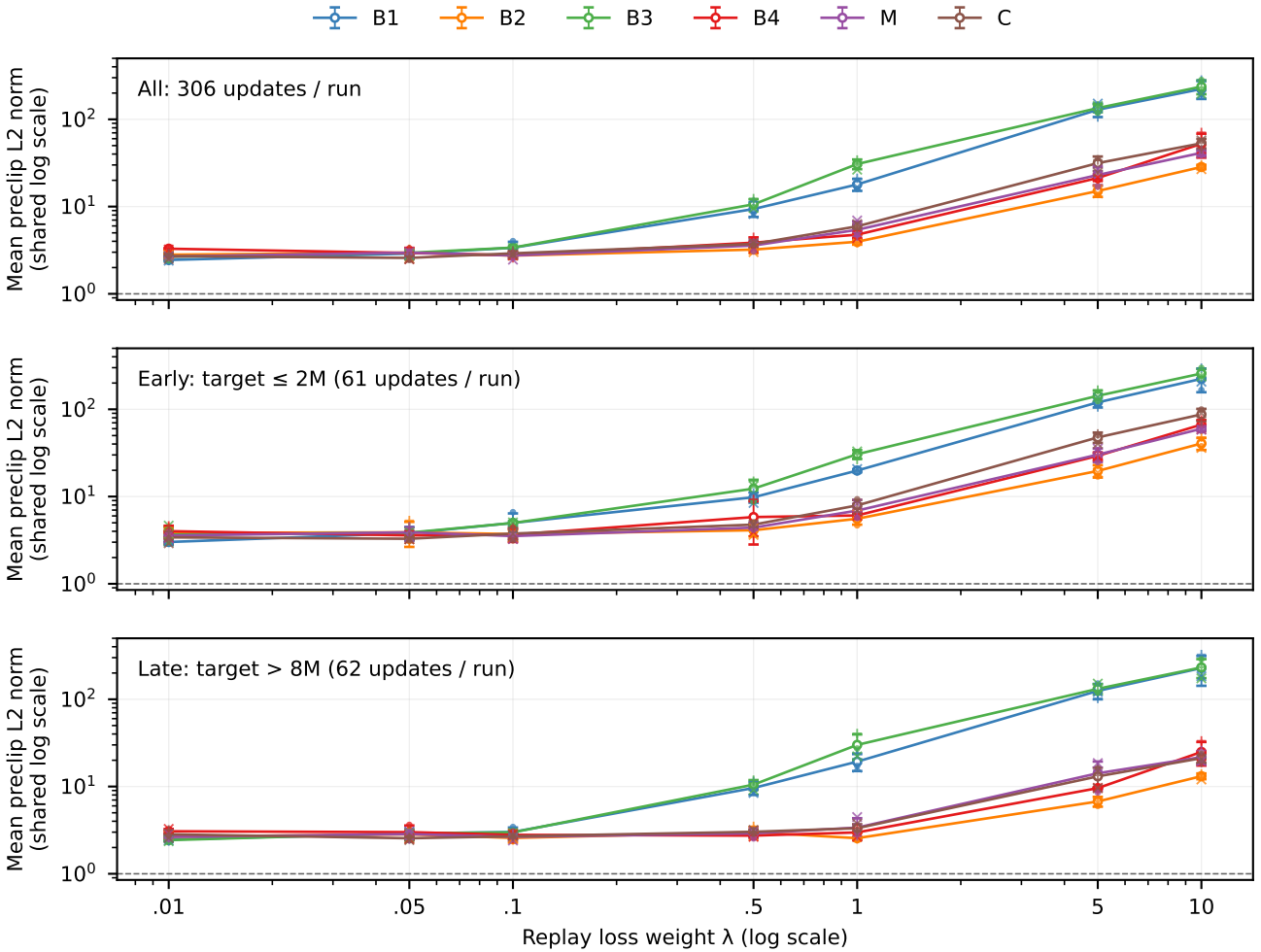
A gradient logs: post-test descriptive diagnostic

ALL 126 runs / 38,556 logged updates / configured threshold = 1
Strict norm > threshold: 100% in ALL / EARLY / LATE for every run
Updates per run: ALL 306 | EARLY (target ≤ 2M) 61 | LATE (target > 8M) 62



All 3 seeds (11: dot; 22: ×; 33: +) and mean ± sample SD; λ/windows share random sources.
Total gradient norm does not determine Adam update magnitude/direction or a causal mechanism.